

Advanced Picking & Multi-Step Workflows – Product Requirements Document

Background

The current Labamu WMS supports batch, cluster and wave picking strategies on the back-end and has a workflow engine that can orchestrate put-away and pick operations via specialised handlers ¹ ² . However, the front-end only exposes a single-order picking session, and the Routes module still defines simple one-step "pull" rules without a graphical way to configure multi-step sequences. The backlog also notes that waveless (continuous-flow) picking is planned but not yet implemented. This gap means warehouse managers cannot take advantage of advanced fulfilment strategies such as batch/cluster/wave picking, QC inspection or staging steps during inbound and outbound workflows, or dynamic routing that adapts to real-time conditions.

Purpose

This PRD defines the functionality required to expose advanced picking strategies and multi-step workflows in the Labamu WMS UI, and to implement waveless continuous flow picking. By closing this gap, operations teams will be able to configure and execute complex fulfilment scenarios – such as wave picking with consolidation, QC inspection before put-away, or staged packing – directly from the user interface. The goal is to ensure parity between the back-end capabilities and the front-end experience, thereby improving productivity, reducing travel time and enabling dynamic re-sequencing based on real-time conditions.

Goals & Objectives

1. **Expose advanced picking strategies in the UI** – Allow operators to start picking sessions with batch, cluster, wave and waveless strategies. Provide controls to group orders by destination, product type, shipping cut-off or other criteria.
2. **Implement waveless (continuous-flow) picking** – Extend the `PickingStrategyService` so it creates continuous-flow sessions where orders are picked as they arrive, with dynamic slot assignment and re-optimisation.
3. **Introduce multi-step routes and workflows** – Replace one-step pull rules with a configurable route editor that allows users to build sequences such as Receive → QC → Put-Away or Wave Pick → QC → Staging → Consolidation → Packing → Shipping. Integrate this editor with the existing workflow engine so steps trigger at the correct time.
4. **Provide UI for dynamic route configuration** – Offer a graphical drag-and-drop editor, similar to the workflow builder, where users can design multi-step routes with optional conditions, QC checkpoints, staging areas and consolidation logic.
5. **Enable dynamic re-sequencing** – Use distance calculations, labour availability and equipment status to re-optimize the sequence of picks and tasks as conditions change. Display suggested re-sequence prompts to supervisors.

Scope

The scope of this enhancement includes:

- Front-end changes to the picking module to allow selection of batch, cluster, wave and waveless strategies.
- Implementation of waveless picking logic in the `PickingStrategyService` and its handler, including continuous assignment of picks as new orders are received.
- Development of a route editor UI within the Warehouse → Routes section that supports multi-step sequences with QC, staging, consolidation and other operational steps.
- Updates to the Routes and Workflow modules to support multi-step routes and to link them to picking and put-away operations through the workflow engine.
- Enhancements to display dynamic routing suggestions and re-sequencing prompts based on real-time data.

The following are out of scope for this phase: 3D visualisation, labour-management modules, multi-client billing, and IoT integrations – these are addressed in separate phases.

Assumptions

- The back-end already supports batch, cluster and wave picking strategies via the `PickingStrategyService` and will be extended to support waveless sessions ¹.
- The workflow engine can trigger put-away and pick tasks through handlers ².
- Roles and permissions are user-defined; access control requirements are not specified here.
- The one-to-many mapping between functional areas and location codes is in place, enabling more granular assignment of picks.

User Stories and Acceptance Criteria

As a	I want	so that	Given	When	Then
Warehouse Manager	to select a picking strategy (batch, cluster, wave or waveless) when starting a picking session	I can optimise picking operations based on order volume and warehouse layout	A draft or reserved sales order exists and the user has the Picking role	I open the Picking module and start a new session, choosing "Batch" as the strategy	The system groups orders into a batch session, creates tasks for each order line and displays the list of pick tasks
Warehouse Manager	to start a waveless (continuous flow) picking session	orders can be picked as they arrive without manual grouping	There are multiple reserved orders, and waveless picking is enabled	I select "Waveless" and click Start	A continuous picking queue is created; new orders are automatically added to the queue and pick tasks are assigned in real time

As a	I want	so that	Given	When	Then
Operations Manager	to configure a multi-step route that includes QC inspection, staging and consolidation steps	quality checks and staging processes can be embedded into inbound/outbound flows	I have access to the Routes module	I open the new route editor, drag steps (Receive, QC Inspect, Put-Away) and define a sequence	The system saves the route, linking each step to its handler; the workflow engine triggers QC and staging tasks when the route executes
QC Inspector	to see tasks generated for QC inspection as part of a route	I can perform inspections before goods are stored or shipped	A multi-step route with a QC step is active	The workflow engine reaches the QC step during execution	A QC inspection task appears in my task list, with fields to enter pass/fail and comments
Picker	to receive picking tasks aggregated by zone when batch or cluster picking is used	I minimise walking distance	A batch or cluster pick session is in progress	I open my mobile picking interface	The system shows my next tasks grouped by location zones, along with quantity and SKU information
Operations Supervisor	to re-optimize pick sequences based on real-time labour availability and congestion data	I can adjust assignments to reduce bottlenecks	Live labour and sensor data are available	I view the pick dashboard during a session	The system highlights suggested re-sequencing (e.g. reassign tasks to a different worker) and I can accept or reject these suggestions

Functional Requirements

1. Picking Strategy Selection UI

- Extend the picking session creation form to include a strategy drop-down with options: *Single*, *Batch*, *Cluster*, *Wave*, and *Waveless*. The default should be *Single* to maintain backward compatibility.
- When *Batch* or *Cluster* is selected, allow the user to set grouping criteria (e.g., destination, carrier, shipping cut-off time).
- When *Wave* is selected, allow configuration of wave size and release cadence.
- When *Waveless* is selected, disable grouping controls and display a notice that tasks will be continuously assigned.

6. Waveless Picking Logic

7. Implement continuous-flow picking in `PickingStrategyService`: upon creation, a waveless session listens for new reserved orders and automatically assigns pick tasks to the queue.
8. Ensure pick tasks created during waveless sessions use real-time distance calculations and labour availability to assign tasks to the nearest available worker.
9. Provide an API endpoint for mobile clients to poll and retrieve the next task in the queue.
10. **Multi-Step Route Editor**
11. Replace the existing one-step route interface with a drag-and-drop editor modelled after the workflow builder. Users should be able to add steps (Receive, QC Inspect, Put-Away, Cross-Dock, Staging, Consolidation, Packing, Shipping) and connect them in sequence.
12. Allow conditional branches (IF/ELSE) to handle scenarios such as damaged goods or cross-dock triggers.
13. Provide a canvas to reorder steps, delete steps, and configure properties (e.g., QC sampling rate, staging area).
14. Persist the route definition as part of the route record and link it to workflow templates for execution.
15. **Integration with Workflow Engine**
16. Map each route step to a workflow step type recognised by the workflow engine (e.g., QC → QC_INSPECT, Staging → STAGE).
17. On execution, the workflow engine should call the appropriate handler (PickHandler, PutawayHandler, QCHandler) to generate tasks.
18. Ensure that route definitions can be activated/deactivated and versioned.
19. **Dynamic Re-sequencing and Monitoring**
20. Build a dashboard for supervisors showing active pick sessions, worker status, and travel distances.
21. Integrate with the floor-plan module to retrieve distances and with labour data to determine worker availability.
22. Implement algorithms to suggest re-sequencing (e.g., reassign tasks to idle workers) and present accept/reject options to supervisors.
23. **UI Enhancements and Accessibility**
24. Ensure the new forms and editors are responsive and accessible for both desktop and mobile users.
25. Provide tooltips and inline help to explain picking strategies and route steps.

Non-Functional Requirements

- **Performance:** Creating a picking session should take no more than 2 seconds. Waveless queues must assign new tasks within 1 second of an order being reserved.
- **Scalability:** The system should handle at least 500 concurrent pick tasks and 20 simultaneous workers without degradation.
- **Reliability:** Route definitions and picking sessions must persist across service restarts. Tasks should be idempotent to prevent duplication.
- **Usability:** Drag-and-drop interactions must be intuitive; user testing should ensure that warehouse operators can configure a multi-step route in under five minutes.

Dependencies

- Extension of `PickingStrategyService` for waveless logic.
- Update to front-end components (React/Vue) to support strategy selection and route editing.
- Integration with the existing workflow engine and handlers for QC, staging and consolidation.
- Availability of real-time labour and distance data from the floor-plan module and, eventually, the labour management module.

Risks & Mitigation

1. **Complexity for users:** Multi-step routes and multiple picking strategies may confuse inexperienced users. Mitigation: provide pre-configured templates and contextual help.
2. **Performance overhead:** Waveless continuous flow and dynamic re-sequencing may increase database and CPU load. Mitigation: optimise queries, cache frequently accessed data and monitor performance under load.
3. **Data consistency:** Ensuring that tasks are correctly created and closed across different steps requires robust transaction management. Mitigation: leverage existing transaction patterns in the workflow engine.

Open Questions

1. Should the grouping criteria for batch/cluster picking be limited to destination, product type and shipping cut-off, or should users be able to define custom fields?
2. How should the system prioritise between wave and waveless sessions when orders could belong to both categories? For example, should urgent orders bypass the current wave and be added to a waveless queue?
3. What level of granularity should the re-sequencing algorithm operate on – entire orders, individual lines or pick tasks? Would users prefer manual overrides or automated adjustments?
4. Are there specific industries (e.g., cold storage, hazardous materials) that require additional route steps or specialised picking modes?

¹ [raw.githubusercontent.com](https://raw.githubusercontent.com/Emmanuelvdg/labamu-wms/main/apps/api/src/workflow/handlers/pick-handler.ts)

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